



Kabul University
Information Science & Engineering Department
Voice-Controlled Virtual Assistant for PwDs (VCVAP)
Full Proposal Application
Deadline for submission of applications:

01/04/2027

SECTION 1: About The Applicant's Organization

Organisation's legal name: Innovation for Tomorrow (IFT)

Organisation's legal status: Active

Application reference number (Number assigned to the concept note): IPwDs-KU-ISE-2026

Year of registration: 2024

Official address: 4th Makrorian, Kabul, Afghanistan

Official website: <https://www.ift.info.af>

Name of the executive director/head of CSO: Shafiullah Haidary

Name of the contact person/s: Sarwar Ebrahimi

Contact telephone: +93 (0) 766337365

Contact email address: contact@ift.info.af

Estimated project budget (AFN/USD): \$1,000,000 (One Million US Dollars)

Project duration (months): 12 months

Proposed start date of the project (day/month/year): 01/04/2026

Proposed end date of the project (day/month/year): 01/04/2027

The location of project site (as specific as possible): 4th Makrorian, 14 Street, IFT Building, Kabul, Afghanistan

Provinces where the lead organisation currently operates: (Please mention all):

Kabul (primary), Herat, Bamyan



Partners: If the project will be implemented with other organisations or groups, provide their details here:

1. Afghan Disability Association (ADA)

Legal status: NGO / Active

Role in the project: Beneficiary liaison, needs assessment, field outreach

2. Ministry of Higher Education, Afghanistan

Legal status: Government Ministry / Active

Role in the project: Policy support and curriculum integration advisory

SECTION 2: Overview Of the Project

Title of the proposed project/project's name:

Voice-Controlled Virtual Assistant for PwDs (VCVAP)

What is the overall project goal?

To design, develop, and deploy an inclusive, locally adapted voice-controlled virtual assistant system that empowers persons with physical, visual, and speech-related disabilities in Afghanistan to access information, communication tools, and digital services independently.

Specific Objectives:

1. Conduct a participatory needs assessment of PwDs across Kabul and two provinces (Bamyan & Herat) within the first 2 months.
2. Design the full system architecture and UI/UX by end of Month 3.5.
3. Develop a functional VCVAP prototype supporting Dari and Pashto speech recognition by Month 6.
4. Complete full backend, frontend, and AI/ML engine integration within the 4-month development phase.
5. Conduct rigorous testing and validation with real PwD users; achieve accessibility compliance by Month 10.5.
6. Deploy and publicly launch the system within Month 11.5; establish ongoing maintenance and support for Month 12.

Target audience/project beneficiaries:

Primary Beneficiaries: Persons with Disabilities (PwDs) including visually impaired, physically disabled, and those with speech impairments.

Expected Numbers: 500 direct beneficiaries (310 men, 140 women, 40 youth aged 14–24, 10 children with disabilities); 2,000+ indirect beneficiaries (families, caregivers, communities).



Funding window applying for:

IT Innovation & Inclusive Technology Fund – Kabul University ISE Research & Development Grant 2026

SECTION 3: Project Description

3.1 RATIONALE & RELEVANCE

Afghanistan has one of the world's highest disability prevalence rates, with an estimated 2.7 million persons with disabilities (PwDs) — nearly 7% of the national population. PwDs face severe barriers in accessing information, education, healthcare, and digital services. These barriers are dramatically amplified by the country's low digital literacy levels, near-absent assistive technology sector, and the complete lack of voice-enabled software in Dari or Pashto.

Despite the global proliferation of voice-controlled assistants (Siri, Google Assistant, Alexa, Cortana), none offer support for Dari or Pashto. This critical linguistic gap renders international assistive technology entirely inaccessible to the vast majority of Afghan PwDs, even when devices are physically available.

The VCVAP project addresses this gap directly. It will produce a locally-designed, culturally appropriate, and linguistically tailored voice-controlled virtual assistant that enables PwDs to independently navigate digital environments, access government services, communicate with families, and engage with educational content — all through natural voice commands in Dari and Pashto.

The project is aligned with SDG 10 (Reduced Inequalities), SDG 4 (Quality Education), and Afghanistan's national disability policy framework. Innovation for Tomorrow (IFT) is uniquely positioned to lead this initiative and active partnerships with the Afghan Disability Association (ADA) and the Ministry of Education.

3.2. METHODOLOGY

The project follows a seven-phase implementation model spanning 12 months of active work:

1. **Phase 1 – Research & Requirements Analysis (2 months):** Participatory needs assessments with PwDs in Kabul, Bamyan, and Herat. User persona development. NLP dataset gap mapping.
2. **Phase 2 – System Design & Planning (1.5 months):** Full system architecture definition. UI/UX wireframing. Database and API design. Accessibility compliance planning (WCAG 2.1).
3. **Phase 3 – Prototype Development (2.5 months):** Core voice recognition engine prototype. Dari & Pashto NLP model. Basic assistant interface.
4. **Phase 4 – Full Development (4 months):** Complete backend API, frontend interface, AI/ML engine integration, cloud hosting, and security hardening.
5. **Phase 5 – Testing & Validation (2 months):** Field testing with 200 real PwD users. Accessibility testing. Bug fixing. Performance and security auditing.
6. **Phase 6 – Deployment & Launch (1 month):** Production deployment. Public launch event. User training for PwDs. Documentation and open-source release.
7. **Phase 7 – Maintenance & Support (1 month):** Ongoing system monitoring, bug fixes, and user support desk.



3.3 MONITORING AND EVALUATION

A dedicated M&E framework will operate throughout all project phases. Monthly progress reports will be submitted to the Project Steering Committee. Key performance indicators include: NLP speech recognition accuracy (>85%), user satisfaction rate (>80%), system uptime (>99%), and number of active PwD users. A baseline assessment in Month 1, mid-term evaluation in Month 6, and final external evaluation in Month 12 will track all log frame indicators disaggregated by gender, disability type, and geography. An online dashboard will provide real-time visibility for all stakeholders.

3.4 SUSTAINABILITY

- **Open-Source Release:** All code will be released under MIT License on a public GitHub repository, enabling free future development.
- **Institutional Embedding:** Ministry of Higher Education will integrate VCVAP into ongoing research programs and student capstone projects.
- **Trained Human Capital:** 50 technicians and 20 support workers will remain operational beyond project closure.
- **Government Partnership:** Formal MOU with the Ministry of Education to integrate VCVAP into public digital services.
- **Revenue Feasibility Study:** Post-project assessment of public-private partnership models for self-sustaining operations.

3.5. CHALLENGES TO THE PROJECT/RISK AND MITIGATION

Challenges include: limited connectivity in rural Afghanistan, absence of Dari/Pashto NLP training data, cultural barriers, security environment, and potential personnel turnover. Each risk has an assigned mitigation strategy and responsible party. The 5% contingency reserve per phase provides financial buffer for escalated risks.

3.6. PROJECT MANAGEMENT

The project is governed by a three-tier structure:

- **Project Steering Committee (PSC):** Innovation for Tomorrow (IFT), ADA, and Ministry of Higher



Education representatives. Meets quarterly. Provides strategic oversight and approves major decisions.

- Project Management Unit (PMU): Sarwar Ebrahimi & Shafiullah Haidari (Co-Project Managers), M&E Officer, Finance Officer, and Field Coordinators. Manages day-to-day operations and submits monthly reports.
- Technical Development Team: Software engineers, AI/ML specialists, UI/UX designer, DevOps engineer, QA testers, and accessibility consultants. Operates under agile sprint methodology.

3.7. LOGICAL FRAMEWORK

Result Level	Description	Indicators	Verification Means	Assumptions
Goal	Improved quality of life & digital inclusion of PwDs	500+ PwDs actively using VCVAP within 12 months	Post-project survey, usage analytics logs	Continued ICT infrastructure availability
Outcome 1	Functional VCVAP system deployed & operational	System passes all accessibility & usability tests; uptime > 95%	Pilot test reports, QA evaluation	Adequate internet and power infrastructure
Outcome 2	Local technical capacity built & sustained	50 technicians trained; 20 support workers certified	Training certificates, attendance sheets	Trained staff remain employed
Output 1	Needs assessment report published	1 comprehensive report by end of Month 2	Published PDF report, stakeholder signoff	PwD community cooperation in surveys
Output 2	System architecture & design finalized	Design documents approved by Month 3.5	Architecture diagrams, design review minutes	Stakeholder agreement on scope
Output 3	Working prototype (Dari & Pashto) completed	Prototype tested by 20 users by Month 6	Demo sessions, prototype test report	NLP training data availability
Output 4	Full system (backend, frontend, AI) integrated	All modules pass integration tests by Month 10	Integration test report, code repository	Skilled developer team retention
Output 5	System publicly launched & deployed	Live system accessible to 500 PwD users by Month 11.5	Deployment report, user registration logs	App store approvals, hosting uptime
Output 6	Maintenance & support system operational	Support response time < 24 hrs; monthly uptime > 99%	Support ticket logs, uptime reports	Budget for ongoing cloud hosting secured



3.8. BUDGET

Summary – All Phases

Phase	Duration	% of Budget	Total (USD)	Status
Phase 1: Research & Requirements Analysis	2 months	6.4%	\$64,000	Planned
Phase 2: System Design & Planning	1.5 months	11.4%	\$114,000	Planned
Phase 3: Prototype Development	2.5 months	16.4%	\$164,000	Planned
Phase 4: Full Development	4 months	41.4%	\$414,000	Planned
Phase 5: Testing & Validation	2 months	11.4%	\$114,000	Planned
Phase 6: Deployment & Launch	1 month	6.4%	\$64,000	Planned
Phase 7: Maintenance & Support	1 month	6.4%	\$64,000	Planned
TOTAL PROJECT BUDGET	13 months	100%	\$1,000,000	Approved

3.9. ANNEXES

[Annex 1.1. Activity Timeline](#)

[Annex 1.2. Project log frame](#)

[Annex 1.3. Project Risk Register](#)

[Annex 1.4. Project Budget](#)

[Annex 1.5. Monitoring plan](#)